DRUGS · SPECIAL ENT AGENTS

Glycopyrronium and hyoscine for drooling

Drooling (sialorrhoea) affects many children with cerebral palsy and neurodisability and adults with Parkinson's disease, motor neurone disease and after stroke. It is usually a problem of poor oral control and swallowing rather than of excess saliva. Anticholinergic drugs reduce saliva production: glycopyrronium by mouth, hyoscine as a skin patch. Treatment sits alongside therapy, botulinum toxin and, for severe cases, surgery to the ducts or glands.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

GLYCOPYRRONIUM AND HYOSCINE FOR DROOLING

	SUMMARY
Glycopyrronium	Quaternary antimuscarinic that does not cross the blood-brain barrier well; oral solution (for example 320 micrograms/ml) for children from 3 years: start about 12.8 micrograms/kg three times daily, increased weekly to a maximum of about 64 micrograms/kg (or 6 ml) three times daily, by product
Hyoscine hydrobromide	Transdermal patch (1 mg over 72 hours) behind the ear; children by age (often half a patch to one patch every 3 days)
How they work	Block muscarinic M3 receptors in the salivary glands, reducing watery saliva
Other options	Speech and language therapy and positioning; botulinum toxin injection; submandibular duct relocation or ligation, gland excision, or ductal ligation (four-duct ligation); trihexyphenidyl in dystonic children

2 CHAPTER TWO Using them in ENT

Practice.

2.1 Assessing drooling

Assess its severity and frequency, posture and head control, dental and oral health, reflux, and aspiration risk (posterior drooling causes chest infections and needs different priorities). Involve the multidisciplinary team.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Constipation, urinary retention, flushing, dry skin and reduced sweating (overheating in hot weather), thickened secretions (harder to clear, a particular concern with posterior drooling), blurred vision, behavioural change and irritability, tachycardia; hyoscine crosses into the brain and can cause confusion and drowsiness, and local skin irritation.

3.2 Contraindications

Glaucoma, bowel obstruction, urinary retention, myasthenia gravis, significant cardiac disease.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Assessment of drooling and the multidisciplinary approach.
- Anticholinergic side effects in children.
- Surgical options when drugs fail.

4.2 Common pitfalls

COMMON PITFALLS

- Thickening secretions in a child who aspirates.

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